

MISAWA AB, Japan

WMO: 475800

Lat: 40.705N

Lon: 141.372E

Elev: 129

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 44402

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.0	20.8	8.9	8.7	24.2	10.8	9.6	25.1	30.3	30.6	28.2	30.1	10.5	280	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	86.1	74.7	82.7	72.9	80.3	72.0	76.6	83.1	75.0	80.2	73.5	78.4	12.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.9	131.6	80.6	73.3	124.6	78.5	71.7	117.8	76.8	40.2	83.2	38.6	79.4	37.2	78.4	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	24.0	21.1	13.8	91.5	3.0	2.8	11.7	93.5	9.9	95.1	8.3	96.7	6.1	98.8
			12.7	79.2	2.7	2.3	10.7	80.8	9.2	82.2	7.7	83.5	5.7	85.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	29.0	30.1	36.4	46.9	55.7	61.2	68.9	72.3	66.6	55.8	44.5	33.9
	DBStd	15.92	4.67	5.24	6.07	6.40	6.12	5.48	6.31	5.35	5.33	5.42	6.75	5.62
	HDD50	2486	652	557	424	135	18	0	0	0	0	11	188	501
	HDD65	5907	1117	977	888	545	295	139	32	5	40	288	615	966
	CDD50	2564	0	0	2	41	194	336	586	691	499	192	23	0
	CDD65	508	0	0	0	1	6	25	153	230	90	3	0	0
	CDH74	2687	0	0	0	8	50	127	867	1278	352	5	0	0
	CDH80	626	0	0	0	1	7	19	206	331	62	0	0	0
Wind	WSAvg	9.5	10.9	10.8	11.4	11.1	9.4	7.9	7.5	7.8	8.2	9.0	9.9	10.5
Precipitation	PrecAvg	46.2	2.6	2.1	2.6	2.8	3.5	4.0	5.6	6.5	4.3	3.3	2.8	
	PrecMax	58.4	5.2	6.0	4.6	5.8	5.6	8.0	10.9	12.9	12.7	8.5	6.4	7.4
	PrecMin	34.9	1.1	0.8	0.8	0.6	1.3	1.3	0.9	1.0	2.1	1.1	1.1	1.4
	PrecStd	6.3	1.1	1.1	1.1	1.3	1.3	1.6	2.5	3.3	2.6	2.2	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	50.3	61.1	73.5	79.2	82.6	89.5	91.1	86.1	73.7	66.0	53.9
		MCWB	39.2	43.3	48.5	55.1	60.1	67.0	76.4	76.5	74.5	63.0	56.9	47.8
	2%	DB	41.0	44.6	55.1	66.4	73.7	77.2	85.7	86.4	80.9	70.0	61.1	49.7
		MCWB	36.3	39.1	45.5	51.9	59.0	65.6	74.3	75.6	71.9	61.6	53.8	44.8
	5%	DB	38.9	41.3	50.3	62.2	70.1	73.7	82.2	84.0	77.4	67.9	58.7	46.3
		MCWB	34.8	35.9	42.4	49.7	57.6	64.4	73.1	74.6	69.5	60.2	52.1	42.0
	10%	DB	35.9	39.0	46.5	57.6	66.4	70.2	78.9	80.7	75.1	64.8	55.5	42.8
		MCWB	32.3	34.3	40.0	47.9	55.6	63.0	71.7	73.2	68.2	58.2	49.7	38.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	44.5	50.3	58.4	64.6	71.2	78.6	79.1	76.1	67.1	59.3	49.8
		MCDB	42.9	49.2	57.5	67.5	72.9	77.5	85.7	86.3	83.0	70.8	62.6	53.0
	2%	WB	37.0	39.4	46.1	54.4	61.6	68.5	76.0	77.3	73.6	63.7	55.3	45.2
		MCDB	39.9	43.5	52.5	62.4	70.1	74.3	82.5	84.0	79.1	67.6	60.3	48.6
	5%	WB	34.8	36.7	43.2	51.5	59.4	66.1	74.3	75.7	71.8	61.8	52.8	41.9
		MCDB	37.5	40.7	49.3	59.1	67.3	71.8	80.1	81.2	76.5	65.9	57.4	45.3
	10%	WB	32.8	34.3	40.5	49.4	57.3	63.8	72.6	74.4	69.7	59.8	50.5	38.7
		MCDB	35.4	38.0	45.3	56.1	64.3	69.0	78.0	79.1	73.3	63.9	54.7	42.2
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.9	13.3	16.4	15.4	12.2	11.0	10.7	12.9	15.1	14.0	10.9
		MCDBR	13.4	15.5	19.5	22.9	22.4	18.6	16.6	15.9	15.1	16.9	18.7	15.8
		MCWBR	10.7	12.0	13.4	12.9	11.5	9.6	8.4	7.4	8.8	11.0	13.2	12.7
	5% WB	MCDBR	12.7	15.1	18.7	19.8	19.0	15.7	14.8	13.6	13.2	14.4	17.2	15.6
		MCWBR	10.9	12.5	14.4	13.2	11.6	9.3	8.8	7.5	8.7	10.8	13.6	13.3
Clear-Sky Solar Irradiance	taub	0.364	0.406	0.476	0.524	0.539	0.520	0.522	0.493	0.437	0.412	0.394	0.374	
	taud	2.078	1.959	1.870	1.835	1.879	2.011	2.079	2.170	2.281	2.225	2.166	2.109	
	Ebn at Noon	239	245	242	240	240	245	243	247	253	244	227	223	
	Edh at Noon	36	47	57	64	62	55	51	45	38	36	33	32	
All-Sky Solar Radiation	RadAvg	486	717	1066	1413	1560	1533	1362	1312	1141	886	571	414	
	RadStd	24	61	77	132	162	118	152	155	93	75	44	29	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Station Only	N/A	N/A	-1.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page